

# SIMEET NAYAN

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## EXPERIENCE

### Software Engineer, Wells Fargo

August, 2024 - Present

- **Engineered end-to-end cryptocurrency trading** functionality for millions of mobile users, developing native iOS UIs (Swift, SwiftUI) and low-latency backends (Java, Spring Boot) to power real-time quote streaming, order execution, market data analytics, and portfolio management.
- **Architected an automated recurring payment recommendation engine** using Python (pandas, scikit-learn) and deployed a custom ranking model into Java backend services to deliver personalized, on-device recommendations at sub-200ms p99 latency.
- **Designed and built a custom client-side fuzzing framework** from scratch, enabling developers to simulate complex response scenarios and systematically validate frontend UI resilience across edge-case runtime conditions.

### Summer Intern, Wells Fargo

May, 2023 - July, 2023

Developed a deep learning model for mortgage prepayment risk prediction, improving accuracy and achieving 10× faster response time, while authoring comprehensive test cases for robust API deployment.

## EDUCATION

### Indian Institute of Engineering Science and Technology, Shibpur

2020-2024

- B. Tech. in Information Technology, CGPA : 9.33/10
- Relevant Courses: Data Structures, Algorithms, Machine Learning, Deep Learning, Operating Systems, Databases, Probability & Statistics, Computer Networks

## PERSONAL PROJECTS / OPEN SOURCE CONTRIBUTIONS

- **VectorSwift** [\[Github\]](#): Vector database in Swift. Exact nearest-neighbor search (L2, inner product, cosine), WAL-backed durability, sealed segments. Currently under active development
- **ODSE: Open Data Science Environment** [\[Github\]](#): Persistent, sandboxed RL environment where agents write Python to explore data, train models, and submit predictions on a hidden test set. OpenEnv-compatible; live on Hugging Face.
- **Training a Deep-Q Network for Breakout** [\[Github\]](#): PyTorch implementation of Mnih et al. Trained a DQN agent on Atari Breakout April, 2025
- **Implementation of HyperGAN in Pytorch** [\[Github\]](#): Implementation of Ratzlaff et al. with an ensemble of LeNets. CIFAR-10 test accuracy is about 0.60 vs about 0.55 for a single LeNet. July, 2024
- **Ray(Anyscale)** [\[PRs\]](#): Contributions to Ray, a framework for scaling distributed machine learning workloads.
- **PyTorch-Ignite** [\[PRs\]](#): Optimized metric handling and CI stability

## SKILLS

- Languages: Python, C++, Java, Swift, SQL, Javascript
- ML: PyTorch, scikit-learn, NumPy, pandas, Reinforcement Learning
- Serving and tools: FastAPI, Kafka, Redis, PostgreSQL, Docker, Git, Linux, CI/CD

## ARTICLES

### Publications

- [MRI Insights](#): Classification of Brain Tumor Using Intelligent Feature Extraction Techniques IEEE, January, 2025

### Blogs:

- Training an Agent to Play Breakout using Deep Reinforcement Learning [\[medium.com\]](#) April, 2025
- Deriving Naive Bayes' From Scratch [\[medium.com\]](#) July, 2021